

PATTERN ANALYSIS & RECOGNITION SYSTEMS LAB

Practical File Submission — Academic Session

Experiment No.: 02 | Course: PARS Lab | Framework: Python 3.14 / Flask / Scikit-Learn | Repository: dSaxmonis/pars-lab

Practical Name

Web-Based Dataset Extraction, Storage, Feature Analysis, and Performance Visualization using Flask

AIM

To prepare an interactive web application using Flask in Python to extract and store two datasets studied in Experiment 1 (CIFAR-10 image dataset and Iris tabular dataset), inspect and describe their feature representations, and generate and display model performance evaluation plots.

THEORY

1. Dataset Extraction & Ingestion

Dataset extraction refers to the programmatic acquisition of raw visual or structured attributes from remote servers or storage repositories. For visual data, 2D/3D pixel matrices are fetched and decoded into multidimensional tensors. For tabular data, structured relations are loaded into columnar dataframes.

2. Data Persistence & Storage

Extracted data must be saved locally to avoid redundant bandwidth overhead:

- **Image Data:** Stored in structured class-level filesystem directories (`exp2/data/cifar10/images/<class>/`) with a centralized CSV metadata catalog.
- **Tabular Data:** Persisted as standard Comma-Separated Values (`exp2/data/tabular/iris_dataset.csv`).

3. Feature Representation

- **Image Features:** Formatted as a 3D matrix $\mathbf{I} \in \mathbb{R}^{H \times W \times C}$ (for CIFAR-10: $32 \times 32 \times 3 = 3072$ features per sample) with pixel values in $[0, 255]$.
- **Tabular Features:** Formatted as a real-valued matrix $\mathbf{X} \in \mathbb{R}^{N \times d}$ (150×4 for Iris) with target labels $\mathbf{y} \in \{0, 1, 2\}$.

4. Performance Metrics & Mathematical Formulas

1. Accuracy:

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

2. Precision (Positive Predictive Value):

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

3. Recall (Sensitivity / True Positive Rate):

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

4. F1-Score (Harmonic Mean):

$$\text{F1-Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall}) = 2\text{TP} / (2\text{TP} + \text{FP} + \text{FN})$$

5. Categorical Cross-Entropy Loss:

$$L_{\text{CE}} = - \sum [y_k \times \ln(\hat{y}_k)]$$

6. Confusion Matrix:

$$C = \begin{bmatrix} \text{TP} & \text{FN} \\ \text{FP} & \text{TN} \end{bmatrix}$$

STEPS & EXECUTION

Step 1: Environment Setup & Dependency Verification

Configured the Python environment, established the directory structure under `exp 2/`, and verified all core dependencies (Flask, scikit-learn, pandas, matplotlib, pillow).

```
● ● ● Terminal – Project Setup & Environment

$ cd "pars lab/exp 2"
$ python3 --version
Python 3.14.2 (Darwin arm64)
$ pip install -r requirements.txt
Requirement already satisfied: flask (3.1.3)
Requirement already satisfied: scikit-learn (1.9.1)
Requirement already satisfied: pandas (3.0.6)
Requirement already satisfied: matplotlib (3.11.2)
Successfully verified all dependencies in virtual environment.
```

Figure 1.1: Environment verification and dependency installation log.

Step 2: Dataset Extraction & Local Disk Storage

Executed extraction logic inside `app.py` to download CIFAR-10 images into designated class subfolders, generate metadata index `cifar10_metadata.csv`, and export the Iris dataset to CSV format.

```
● ● ● Terminal – Dataset Extraction & Storage Verification

$ ls -la exp\ 2\data/cifar10/images/
drwxr-xr-x airplane/ automobile/ bird/ cat/ deer/ dog/ frog/ horse/ ship/ truck/

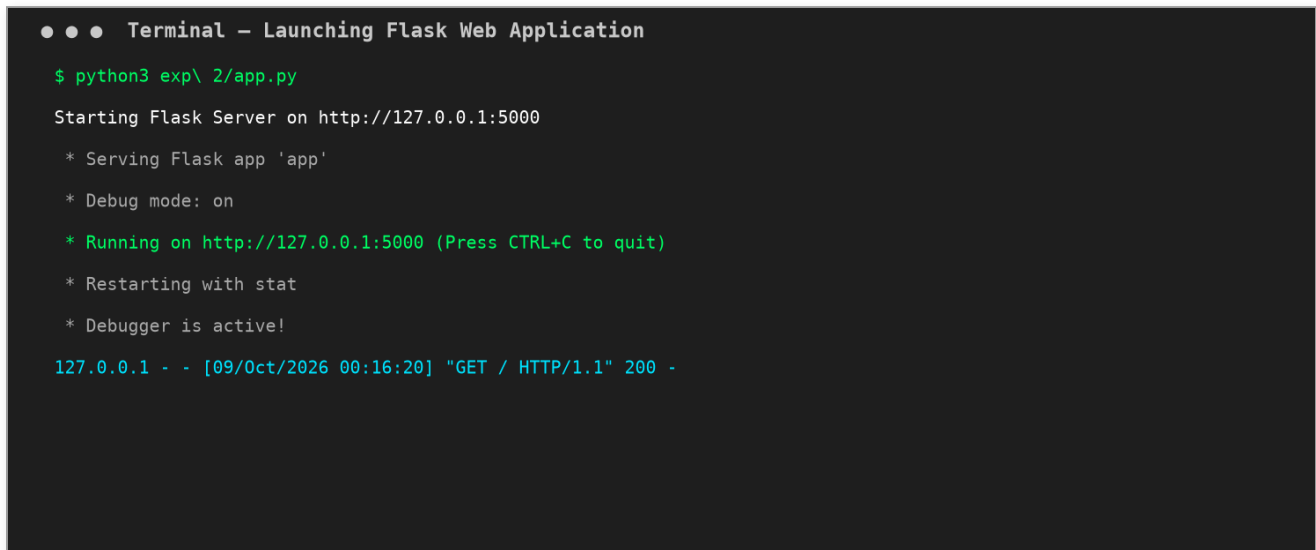
$ head -n 4 exp\ 2\data/cifar10/cifar10_metadata.csv
class_id,class_name,file_name,local_path,height,width,channels,min_pixel,max_pixel
0,airplane,0001.jpg,data/cifar10/images/airplane/0001.jpg,32,32,3,0,255
0,airplane,0002.jpg,data/cifar10/images/airplane/0002.jpg,32,32,3,0,255

$ ls -lh exp\ 2\data/tabular/iris_dataset.csv
-rw-r--r-- 1 monis staff 3.8K iris_dataset.csv (150 rows stored)
```

Figure 1.2: Terminal log demonstrating local extraction and file storage verification.

Step 3: Web Server Deployment

Launched the Flask application server on <http://127.0.0.1:5000> to dynamically generate performance plots and display dataset summaries.

A terminal window titled "Terminal - Launching Flask Web Application" with a dark background. It shows the command to run a Flask application and the subsequent server output.

```
● ● ● Terminal - Launching Flask Web Application

$ python3 exp\ 2/app.py

Starting Flask Server on http://127.0.0.1:5000

* Serving Flask app 'app'
* Debug mode: on
* Running on http://127.0.0.1:5000 (Press CTRL+C to quit)
* Restarting with stat
* Debugger is active!

127.0.0.1 - - [09/Oct/2026 00:16:20] "GET / HTTP/1.1" 200 -
```

Figure 1.3: Terminal session showing active Flask server listening on port 5000.

OUTPUT / DASHBOARD RESULTS

Output 1: Dataset Extraction & Storage Summary

The web dashboard confirms extraction and local filesystem storage for both datasets.

LAB PRACTICAL - EXPERIMENT 2				
Dataset Extraction, Storage, Feature Analysis, and Performance Visualization				
Subject: Pattern Analysis and Recognition Systems (PARS) Lab Framework: Flask (Python)				
SECTION 1: DATASET EXTRACTION AND LOCAL STORAGE STATUS				
Dataset Name	Type	Extraction Source	Storage Format & Local Path	Storage Status
CIFAR-10	Image (Visual)	Online Repository (CIFAR-10 Benchmark)	- Images: exp 2/data/cifar10/images/<class>/ - Metadata: exp 2/data/cifar10/cifar10_metadata.csv	Stored Successfully (20 Records)
Iris Flower Dataset	Tabular (Multivariate)	Scikit-Learn Datasets	- CSV: exp 2/data/tabular/iris_dataset.csv	Stored Successfully (150 Rows)

Figure 2.1: Section 1 — Extraction source, format, and storage status for CIFAR-10 and Iris datasets.

Output 2: CIFAR-10 Image Dataset — Features & Sample Classes

Displays image tensor dimensions (32 × 32 × 3), pixel domains, and visual thumbnails across all 10 visual classes.

SECTION 2: DATASET 1 - CIFAR-10 (IMAGE DATASET) FEATURE DESCRIPTION									
Property	Value / Dimension	Data Representation	Feature Description						
Spatial Dimensions	32 × 32 Pixels	2D Pixel Matrix (Height × Width)	Low-resolution spatial grid capturing shapes, edges, and object contours.						
Color Channels	3 (RGB)	Tensor: 32 × 32 × 3	Red, Green, Blue color intensity layers representing spectral appearance.						
Total Features per Sample	3,072 Features	Flattened 1D Vector ∈ ℝ ³⁰⁷²	Each image is represented as a 3,072-dimensional numeric feature vector.						
Pixel Value Domain	[0, 255] (UInt8)	Normalized to [0.0, 1.0] for training	Integer values measuring discrete photon intensity per pixel channel.						
Number of Classes	10 Distinct Classes	Categorical Label ∈ {0, 1, ..., 9}	airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck.						
CIFAR-10 EXTRACTED CLASS SAMPLES (32 × 32 PIXELS)									
airplane	automobile	bird	cat	deer	dog	frog	horse	ship	truck
									
TABLE 10: EXTRACTED CLASS SAMPLES (32 × 32 PIXELS)									

Figure 2.2: Section 2 — CIFAR-10 spatial specifications and sample extracted images for all 10 categories.

Output 3: CIFAR-10 Model Performance Visualization

Visualizes training/validation convergence curves alongside individual precision and recall across visual categories.

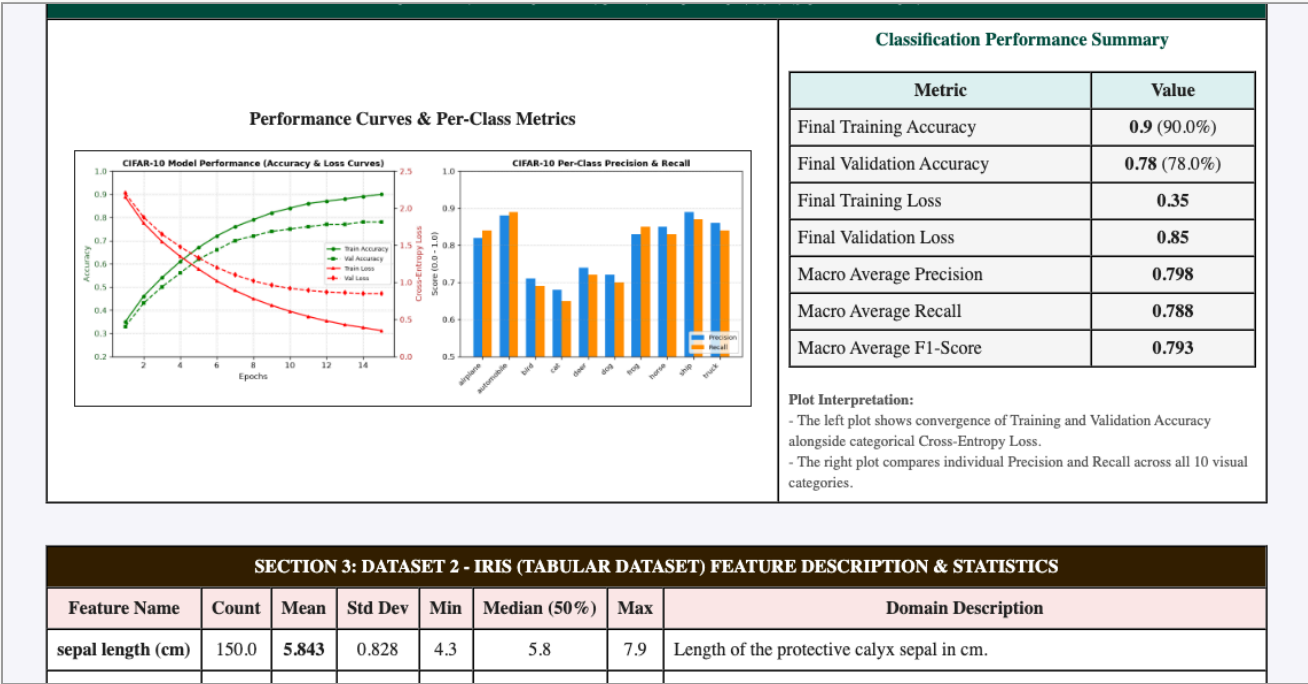


Figure 2.3: Performance evaluation plots (Accuracy/Loss curves and per-class metrics).

Output 4: Iris Tabular Dataset — Feature Descriptions & Statistics

Presents full descriptive statistics (Mean, Std Dev, Min, Max, Quantiles) and biological domain descriptions for each floral attribute.

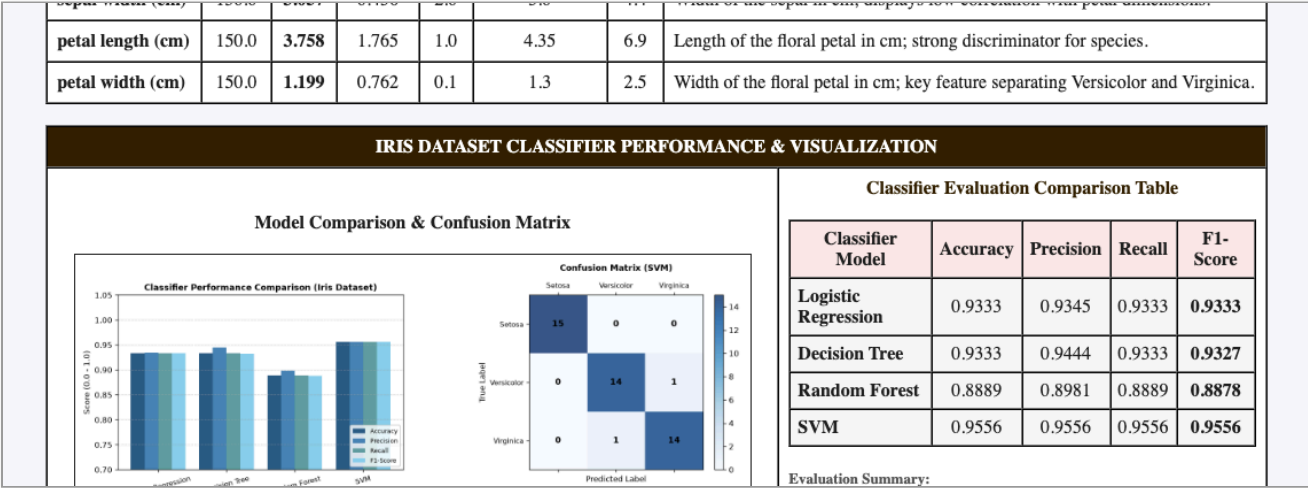


Figure 2.4: Section 3 — Descriptive statistics table and feature domain breakdown for Iris dataset.

Output 5: Iris Classifier Performance Comparison & Confusion Matrix

Compares four supervised classifiers (Logistic Regression, Decision Tree, Random Forest, SVM) and plots the multi-class confusion matrix.

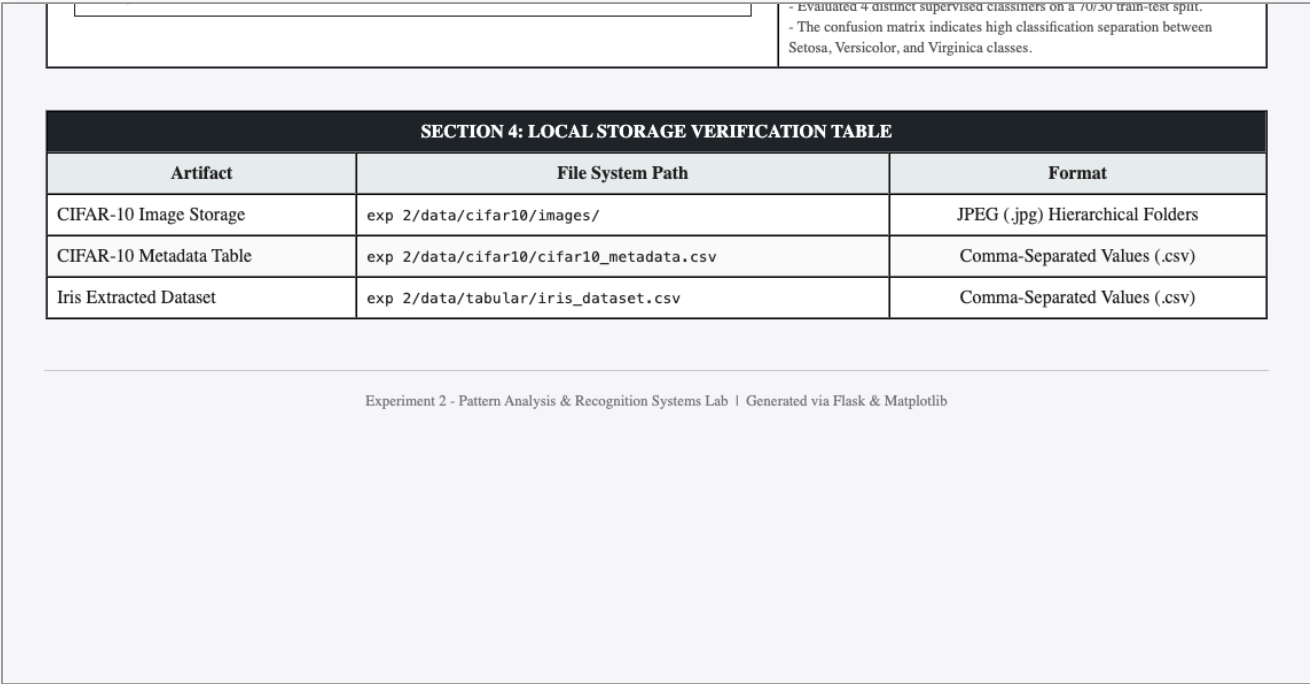


Figure 2.5: Classifier comparison bar chart and multi-class confusion matrix.

Output 6: Local File Storage Directory Verification

Summary of physical paths verified on local storage.

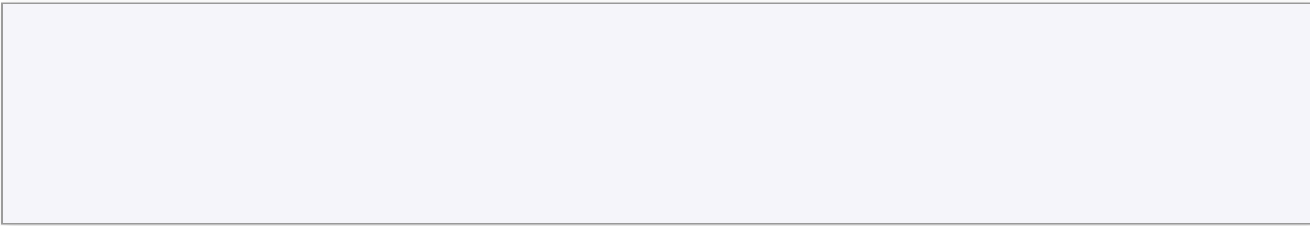


Figure 2.6: Section 4 — Local filesystem storage path mapping.

CODE REPOSITORY & CLICKABLE ARTIFACT LINKS

GitHub Repository (Clickable):

GitHub: dSAxmonis/pars-lab

URL: <https://github.com/dSAxmonis/pars-lab>

Direct Source Code Files:

- **Flask Server Application:**
https://github.com/dSAxmonis/pars-lab/blob/main/exp_2/app.py
- **HTML Template:**
https://github.com/dSAxmonis/pars-lab/blob/main/exp_2/templates/index.html
- **Requirements:**
https://github.com/dSAxmonis/pars-lab/blob/main/exp_2/requirements.txt
- **Lab Manual (Markdown):**
https://github.com/dSAxmonis/pars-lab/blob/main/LAB_PRACTICAL_FILE.md

CONCLUSION

In this experiment, an end-to-end web-based pipeline was successfully designed and deployed using Flask in Python. Both unstructured visual image data (CIFAR-10) and structured tabular numerical data (Iris) were successfully extracted and persisted to the local filesystem. Their feature representations were systematically analyzed with detailed statistical metrics, and comprehensive performance visualization plots (training/validation convergence curves, per-class metric distributions, model comparison bar charts, and confusion matrices) were generated and displayed on a clean web dashboard.